



National University

of Computer & Emerging Sciences Peshawar Campus



Name: Arsem Ali

Roll No: 23P-0038

Program: BCS/BAI/BSE

Semester: Fall – 2023

Time Allowed: 1 hour

Course: NS1001 Applied Physics/Physics for engineers

Examination: Midterm 1

Total Marks: 15 Weightage: 15

Date: Sept 25, 2023

Instructor: Engr M Asif Khan

NOTE: Attempt all questions

Question 1

[5M]

The position of a particle moving along the x-axis varies in time according to the expression $x=3t^2$, where x is in meters and t is in seconds. Evaluate its:

- Position at $t=3s$
- Position at $t = 3s+\Delta t$
- Velocity at $t=3s$

Question 2

[5M]

A jet plane comes in for a landing with a speed of 100m/s and its acceleration can have a maximum magnitude of $5m/s^2$ as it comes to rest.

- From the instant the plane touches the runway, what is the minimum time interval needed before it can come to rest.
- Can this plane land on a small tropical island airport where the runway is 0.900km long? Justify your answer with proper mathematical derivation.

Question 3

[5M]

For fig1 draw the acceleration-time graph.

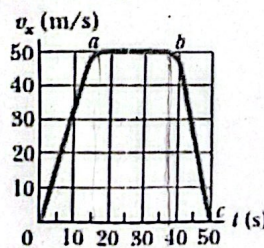


Fig1
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